

MINGBAO (MICHAEL) HE

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EDUCATION

- Expected 12/25 **NEW YORK UNIVERSITY** New York, NY
The Courant Institute of Mathematical Sciences
M.S. in Mathematics in Finance
- **Coursework:** object-oriented programming (Python), algorithmic trading, Black-Scholes model, VaR, covariance matrix estimation, Monte Carlo simulation, data-driven models
- 09/20 - 06/24 **UNIVERSITY OF TORONTO** Toronto, Canada
Bachelor of Science of Mathematics and Its Applications (Probability Stats)
- **Coursework:** machine learning, linear algebra, real analysis, time series
 - **Honors/Awards:** Dean's list, graduated with distinction

EXPERIENCE

- 06/23 - 09/23 **ALLIANZ LIFE INSURANCE** Shanghai, China
Leadership Development Intern
- Developed comprehensive scientific report detailing key factors influencing life expectancy, using linear regression models and gave visual presentation of report
 - Drafted reports, evaluated products, and trained with portfolio manager on investment management fundamentals; gained general understanding of insurance workflow
 - Collaborated with senior management to enhance communication among functional groups, identifying inefficiencies and implementing strategies for improved accountability and efficiency
- 01/23 - 05/23 **HAMILTON HEALTH CENTER** Hamilton, Canada
Data Analyst Intern (Python, SQL)
- Implemented and managed data cleaning pipeline; improving data quality by 55% and significantly increasing efficiency of data extraction process
 - Visualized interactive dashboard of complex product and service data for leaders' critical decision-making; received highly positive feedback from supervisors for process innovation
 - Conducted in-depth statistical analyses on patient data; collaborated with team to identify patterns and trends in health outcomes

PROJECTS

- 09/24 - 12/24 **NEW YORK UNIVERSITY** New York, NY
Performance Analysis of Hedge Fund Returns using Linear Regression (Python)
- Conducted regression analysis on hedge fund returns using the Fama-French 5-Factor model, validating results through t-tests, F-tests to evaluate factor significance and model fit
 - Enhanced predictive accuracy by implementing Elastic Net regularization and expanding regressors with non-linear transformations
 - Derived actionable insights on the key drivers of hedge fund performance, linking model results to economic and financial interpretations
- 09/24 - 12/24 **NEW YORK UNIVERSITY** New York, NY
Machine Learning For Financial Market (Python)
- Built SVR, Decision Trees, and Random Forests to predict Philips (PHG) stock prices and S&P 500 market movements, leveraging technical indicators (EMA, ATR) for feature engineering
 - Applied hyperparameter tuning (GridSearchCV) and cross-validation to optimize model performance, reducing overfitting and improving predictive accuracy

COMPUTATIONAL SKILLS / OTHER

Programming Languages: Python, R, SQL, LaTeX

Languages: English (fluent), Mandarin (native)

Activities: VP of Events of Chinese Undergraduate Students' Associations at University of Toronto